

Instagram Reel Trend Engagement Analysis

Complete Python NumPy Solution

This document contains the complete solution for all six module-level NumPy functions in the Instagram Reel Trend Engagement Analysis task.

1. Import NumPy

```
import numpy as np
```

2. Operation 1: Create Views Array

```
def create_views_array(views_data: list) -> np.ndarray:  
    return np.array(views_data, dtype=np.int64)
```

3. Operation 2: Validate Views Array

```
def validate_views_array(views_array: np.ndarray) -> bool:  
    if views_array.size == 0:  
        return False  
  
    if not np.issubdtype(views_array.dtype, np.number):  
        return False  
  
    return bool(np.all(views_array >= 0))
```

4. Operation 3: Compute View Metrics

```
def compute_view_metrics(views_array: np.ndarray) -> tuple:  
    total_views = int(np.sum(views_array))  
    average_views = round(float(np.mean(views_array)), 2)  
    maximum_views = int(np.max(views_array))  
  
    return total_views, average_views, maximum_views
```

5. Operation 4: Categorize Trend Levels

```
def categorize_trend_levels(views_array: np.ndarray) -> np.ndarray:  
    conditions = [  
        views_array < 3000,  
        (views_array >= 3000) & (views_array < 5000),  
        views_array >= 5000  
    ]  
  
    choices = [  
        "Low Trend",  
        "Moderate Trend",  
        "Viral Trend"  
    ]  
  
    return np.select(conditions, choices)
```

6. Operation 5: Find Longest Growth Streak

```
def longest_growth_streak(views_array: np.ndarray) -> int:  
    if views_array.size == 0:  
        return 0  
  
    if views_array.size == 1:  
        return 1  
  
    current_streak = 1  
    longest_streak = 1  
  
    for i in range(1, len(views_array)):  
        if views_array[i] > views_array[i - 1]:  
            current_streak += 1  
        else:  
            current_streak = 1  
  
    longest_streak = max(longest_streak, current_streak)
```

```
return longest_streak
```

7. Operation 6: Format View Counts

```
def format_view_counts(views_array: np.ndarray) -> np.ndarray:  
    return np.array([f"{int(value):,}" for value in views_array])
```

Complete Code for solution.py

```
import numpy as np

# Operation 1: Create Views Array
def create_views_array(views_data: list) -> np.ndarray:
    return np.array(views_data, dtype=np.int64)

# Operation 2: Validate Views Array
def validate_views_array(views_array: np.ndarray) -> bool:
    if views_array.size == 0:
        return False

    if not np.issubdtype(views_array.dtype, np.number):
        return False

    return bool(np.all(views_array >= 0))

# Operation 3: Compute View Metrics
def compute_view_metrics(views_array: np.ndarray) -> tuple:
    total_views = int(np.sum(views_array))
    average_views = round(float(np.mean(views_array)), 2)
    maximum_views = int(np.max(views_array))

    return total_views, average_views, maximum_views

# Operation 4: Categorize Trend Levels
def categorize_trend_levels(views_array: np.ndarray) -> np.ndarray:
    conditions = [
        views_array < 3000,
        (views_array >= 3000) & (views_array < 5000),
        views_array >= 5000
    ]

    choices = [
        "Low Trend",
        "Moderate Trend",
        "Viral Trend"
    ]

    return np.select(conditions, choices)

# Operation 5: Find Longest Growth Streak
def longest_growth_streak(views_array: np.ndarray) -> int:
    if views_array.size == 0:
        return 0

    if views_array.size == 1:
        return 1

    current_streak = 1
    longest_streak = 1

    for i in range(1, len(views_array)):
        if views_array[i] > views_array[i - 1]:
            current_streak += 1
        else:
            current_streak = 1

        longest_streak = max(longest_streak, current_streak)

    return longest_streak

# Operation 6: Format View Counts
def format_view_counts(views_array: np.ndarray) -> np.ndarray:
    return np.array([f"{int(value):,}" for value in views_array])
```

Sample Input and Expected Results

```
views = np.array([1200, 3500, 1800, 5200, 7600])

validate_views_array(views)
# True
```

```
compute_view_metrics(views)
# (19300, 3860.0, 7600)

categorize_trend_levels(views)
# ["Low Trend", "Moderate Trend", "Low Trend",
#  "Viral Trend", "Viral Trend"]

longest_growth_streak(views)
# 3

format_view_counts(views)
# ["1,200", "3,500", "1,800", "5,200", "7,600"]
```